Signals and Systems

Formulae Standard Laplace Transforms and z-Transforms

Signal	Transform	ROC
$u(t)$	$\frac{1}{s}$	$\mathrm{Re}(s) > 0$
$tu(t)$	$\frac{1}{s^2}$	$\mathrm{Re}(s) > 0$
$\delta(t-\tau),\ \tau \geq 0$	$e^{-s\tau}$	$\forall s$
$e^{-at}u(t)$	$\frac{1}{s+a}$	$\mathrm{Re}(s) > -a$
$te^{-at}u(t)$	$\frac{1}{(s+a)^2}$	$\mathrm{Re}(s) > -a$
$\cos(w_1 t)u(t)$	$\frac{s}{s^2+w_1^2}$	$\mathrm{Re}(s) > 0$
$\sin(w_1 t)u(t)$	$\frac{w_1}{s^2+w_1^2}$	$\mathrm{Re}(s) > 0$
$e^{-at}\cos(w_1 t)u(t)$	$\frac{s+a}{(s+a)^2+w_1^2}$	$\mathrm{Re}(s) > -a$
$e^{-at}\sin(w_1 t)u(t)$	$\frac{w_1}{(s+a)^2+w_1^2}$	$\mathrm{Re}(s) > -a$
$x[n-n_0]$	$z^{-n_0}X(z)$	$z \in R_x$ but may exclude $z = 0,\ \infty$
$\alpha^n u[n]$	$\frac{1}{1-\alpha z^{-1}}$	$\lvert z\rvert > \lvert\alpha\rvert$

1. *This question caries 40% of the marks.*

a) Consider the signal

$$x(t) = \begin{cases} t & 0 \le t \le 1 \\ 0 & \text{otherwise.} \end{cases}$$

For each of the following signals $y(t)$, find an expression for $y(t)$ not involving x. Draw illustrative sketches to show the sequence of steps that must be taken to form $y(t)$ from $x(t)$.

i) $y(t) = x(2t + 1.5)$ [3]

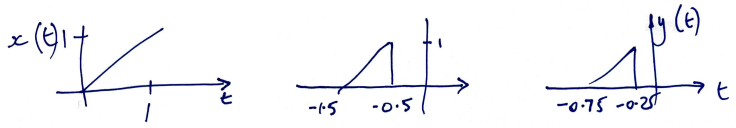
ii) $y(t) = x(-3t - 7)$ [3]

Solution:

(i)

$$y(t) = \begin{cases} 2t + 1.5 & -0.75 \le t \le -0.25 \\ 0 & \text{otherwise} \end{cases}$$

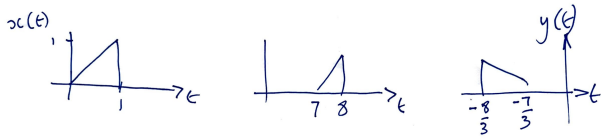
$y(t)$ is formed from $x(t)$ by:



(ii)

$$y(t) = \begin{cases} -3t - 7 & -\frac{8}{3} \le t \le -\frac{7}{3} \\ 0 & \text{otherwise} \end{cases}$$

$y(t)$ is formed from $x(t)$ by:



[*2 marks for expression, 1 mark for sketches*]

b) i) Define a memoryless system. [3]

ii) The signal $y[n]$ is the output of an LTI system $H(z)$ when the input to the systems is $x[n]$ and

$$y[n] = 0.71(x[n] - 2.29x[n-1] - 0.46y[n-2] + 7.35).$$

Is $H(z)$ a memoryless system?

- If 'yes', give a quantified justification in terms of the properties of $H(z)$.
- If 'no', quantify how much memory is used.

[3]

Solution:
(i) A system is memoryless iff the output signal of a system is a function of only the value of the input signal at the present time so that there is no dependence on past (or future) signal values.
(ii) $H(z)$ is not memoryless and needs to store 4 samples.

c) State whether the following signals are causal or noncausal, periodic or nonperiodic, and even, odd or non-symmetric. If periodic, state the period.

i) $x(t) = |A\sin(2\pi t)|$ where A is a scalar constant. [2]

Solution:
non-causal, periodic with period $T = 0.5$ s, even.

ii) $x(t) = 7e^{2(t+1)}$ [2]

Solution:
noncausal, nonperiodic, nonsymmetric

d) Given the frequency spectrum

$$X(j\omega) = 10\delta(\omega - 3),$$

find the corresponding time domain signal, $x(t)$, using the inverse Fourier transform. [4]

Solution:
$x(t) = \frac{5}{\pi}e^{j3t}$

e) Consider the discrete-time signal

$$x[n] = \cos(n\frac{\pi}{3} + \phi).$$

i) State the period of this signal and the corresponding units. [2]

ii) Find the coefficients $X[k]$ of the DTFS of $x[n]$. [4]

Solution:
(i) Period is 6; units of samples.
(ii) Write

$$x[n] = \frac{1}{2}e^{j\frac{\pi}{3}n}e^{j\phi} + \frac{1}{2}e^{-j\frac{\pi}{3}n}e^{-j\phi}.$$

Write the definition of the DTFS for $N = 6$

$$\begin{aligned} x[n] &= \sum_{k=-2}^{3} X[k]e^{jk\frac{\pi}{3}n} \\ &= X[-2]e^{-j2\frac{\pi}{3}n} + X[-1]e^{-j\frac{\pi}{3}n} + X[0] \\ &\quad + X[1]e^{j\frac{\pi}{3}n} + X[2]e^{j2\frac{\pi}{3}n} + X[3]e^{j3\frac{\pi}{3}n}. \end{aligned}$$

Comparing terms at the same frequency gives

$$\begin{aligned} \text{For } k=-1: &\quad X[-1]e^{-j\frac{\pi}{3}n} = \frac{1}{2}e^{-j\frac{\pi}{3}n}e^{-j\phi} \Longrightarrow X[-1] = \frac{1}{2}e^{-j\phi}. \\ \text{For } k=1: &\quad X[1]e^{j\frac{\pi}{3}n} = \frac{1}{2}e^{j\frac{\pi}{3}n}e^{j\phi} \Longrightarrow X[1] = \frac{1}{2}e^{j\phi}. \\ \text{For other } k: &\quad X[k] = 0. \end{aligned}$$

f) Consider the discrete-time signal

$$x[n] = 7\lambda^n e^{jn\Omega_0}u[n]$$

where $u[n]$ is the unit step function and λ and Ω_0 are constants. Find the z-transform of $x[n]$ and its Region Of Convergence. [4]

Solution:

$$X(z) = \frac{7}{1-\lambda e^{j\Omega_0}z^{-1}}$$

with ROC given by $|\lambda e^{j\Omega_0}z^{-1}| < 1$ so that $|z| > |\lambda|$ since $|e^{j\Omega_0}| = 1$.

g) For the last few hundred years, astronomers have kept an annual record of the number of observed solar sunspots, known as the Zurich number. These observations reveal a pattern in solar activity over several years.

The yearly evolution of the Zurich number between the years 1700 and 1987 is plotted in Fig. 1.1. The power spectrum of this signal, obtained using a 288-point DFT, is plotted in Fig. 1.2.

i) What is the sampling frequency of this signal in Hz? [1]

Solution: $(1/(60*60*24*365.25)) = 3.1688 \times 10^{-8}$ Hz, or once per year. Also accept 3.1710×10^{-8} Hz.

ii) Do you expect the result of the DFT to be real-valued, purely imaginary, or complex-valued? Justify your answer. [2]

Solution: Complex, no symmetry in the time-domain signal.

iii) Prior to taking the DFT, the time-domain signal was normalised by removing its mean. What does this step achieve from the point of view of the frequency domain? What would the spectrum look like without this step? [2]

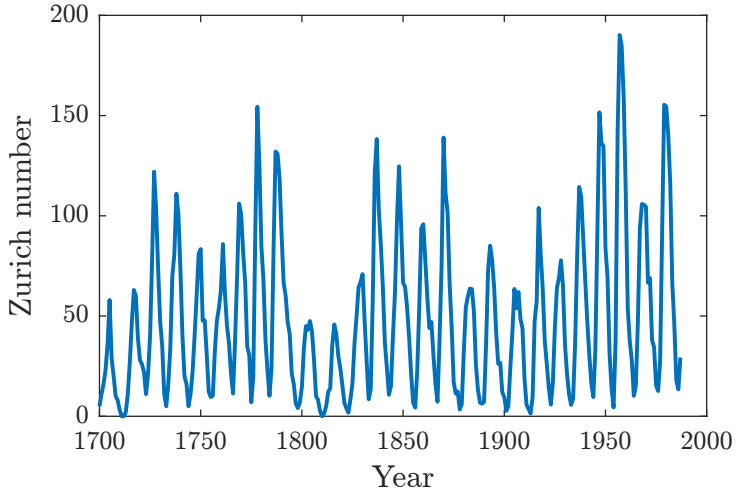


Figure 1.1

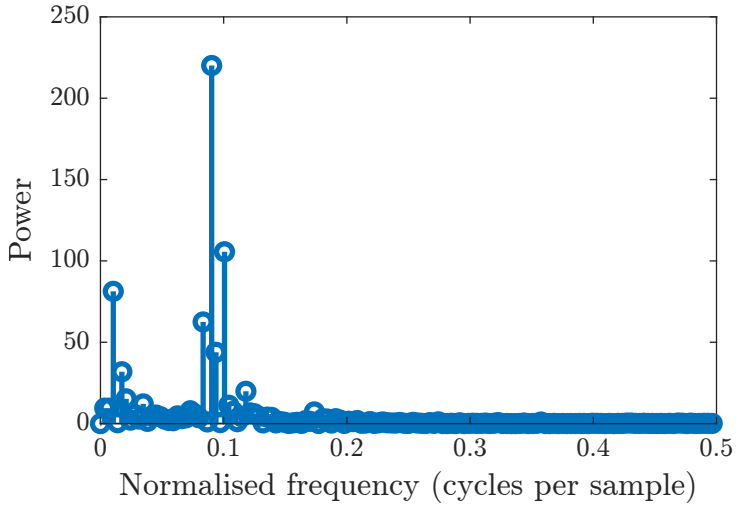


Figure 1.2

Solution: Sets DC component to zero. Without it, the spectrum would contain a large peak at 0 Hz.

iv) Give the period of the solar cycle and explain how you obtained it. [3]

Solution: Approx. 10 years. This can be obtained using the peak at 0.1 in DFT or the distance between high frequency peaks in the time domain.

v) Imagine the solar period itself varies every 100 years. How would you proceed to identify the solar period in every century? (*Hint: consider employing a similar approach to that used for the changing piano notes exercise in the coursework*). [2]

Solution: Take 3 different 100-point DFTs over different periods and identify the peak in each.

2. *This question caries 30% of the marks.*

a) Let $H(s)$ be a causal LTI continuous-time system with input $x(t)$ and output $y(t)$.

i) Write down the expression for the system's impulse response, $h(t)$, in terms of $H(s)$. [2]

ii) Write down the expression involving s for a general complex exponential signal. [1]

iii) Given that $x(t)$ is a particular complex exponential with $s = s_0$, prove that the output $y(t) = Ax(t)$ for some constant A, and write an expression for A in terms of H. [5]

iv) For a particular stable system $H(s)$, the impulse response is

$$h(t) = \gamma e^{-4t}u(t) + \lambda e^{-2t}u(t)$$

where γ and λ are constants. It is known that:

$$y(t) = \frac{11}{15}e^{t} \quad \text{when } x(t) = e^{t} \text{ and}$$
$$y(t) = \frac{7}{12}e^{2t} \quad \text{when } x(t) = e^{2t}.$$

Find $H(s)$, the Region of Convergence of $H(s)$ and the values of γ and λ. [6]

Solution:
(i) The continuous-time impulse response $h(t)$ can be obtained from $H(s)$ using the inverse Laplace transform

$$h(t) = \frac{1}{2\pi j}\int_{\sigma-j\infty}^{\sigma+j\infty} H(s)e^{st}ds$$

(ii) e^{st}
(iii) Employing convolution to obtain the output

$$y(t) = \int_{-\infty}^{+\infty} x(t-\tau)h(\tau)d\tau = \int_{-\infty}^{+\infty} e^{s_0(t-\tau)}h(\tau)d\tau$$
$$= e^{s_0 t}\int_{-\infty}^{+\infty} e^{-s_0\tau}h(\tau)d\tau = e^{s_0 t}H(s_0) = H(s_0)x(t)$$

and so A is given by

$$H(s) = \int_{-\infty}^{+\infty} e^{-s\tau}h(\tau)d\tau$$

evaluated at $s = s_0$.
(iv) Starting from $h(t)$, the Laplace transform gives

$$H(s) = \frac{\gamma}{(s+4)} + \frac{\lambda}{(s+2)}.$$

and the RoC for stable causal systems requires $\Re(s) > -2$. Then when $s = 1$ we are told

$$\frac{\gamma}{5} + \frac{\lambda}{3} = \frac{11}{15}$$

and when $s = 2$ we are told

$$\frac{\gamma}{6} + \frac{\lambda}{4} = \frac{7}{12}.$$

Solving gives $\gamma = 2$ and $\lambda = 1$.
Hence

$$H(s) = \frac{2}{s+4} = \frac{1}{s+2} = \frac{3s+8}{(s+4)(s+2)} \quad \Re(s) > -2.$$

b) A signal $x(t)$ is applied as the input to a continuous-time LTI system described by its system function

$$H(s) = \frac{s}{s^2 - 5s + 6}.$$

The signal $y(t)$ is the output signal from this system.

i) Find the differential equation relating $y(t)$ to $x(t)$. [4]

ii) Sketch $H(s)$ on the s-plane. [3]

iii) Can the system $H(s)$ be stable and causal? Explain your answer. [2]

iv) Find the impulse response of this system, denoted $h_s(t)$, if the system is stable. [3]

v) Find the impulse response of this system, denoted $h_c(t)$, if the system is causal. [2]

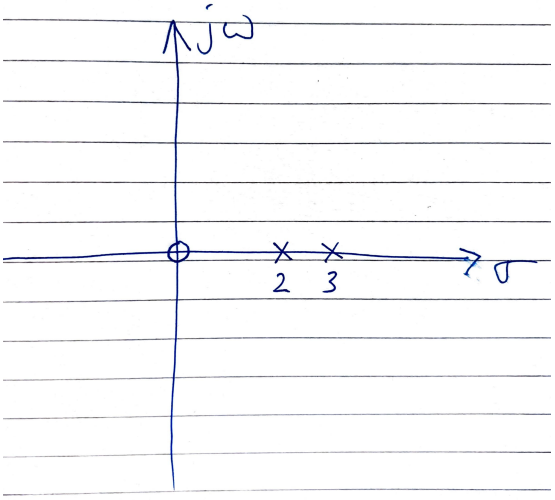
vi) Compute the value of the causal system's impulse response, $h_c(t)$, for $t = -10$ s and $t = 10$ s. [2]

Solution:

(i)

$$\frac{d^2y(t)}{dt^2} - 5\frac{dy(t)}{dt} + 6y(t) = \frac{dx(t)}{dt}$$

(ii) The sketch requires the roots to be determined. These are one zero at the origin and poles at $s = 2$ and $s = 3$.



(iii) No. For a stable and causal system, the poles must be on the left-hand half-plane of s.

(iv) For a stable system, the RoC must include the $j\omega$ axis and therefore lies to the left of the poles, and the impulse response is therefore the sum of two left-sided terms:

$$h_s(t) = 2e^{2t}u(-t) - 3e^{3t}u(-t).$$

(v) For the system to be causal, the impulse response is the sum of only right-sided terms:

$$h_c(t) = -2e^{2t}u(t) + 3e^{3t}u(t).$$

(vi) 0 and $3.2x10^{13}$.

3. *This question caries 30% of the marks.*

a) Consider the continuous-time signal $x(t)$ for which $|X(j\omega)|$ is shown in Fig. 3.1 .

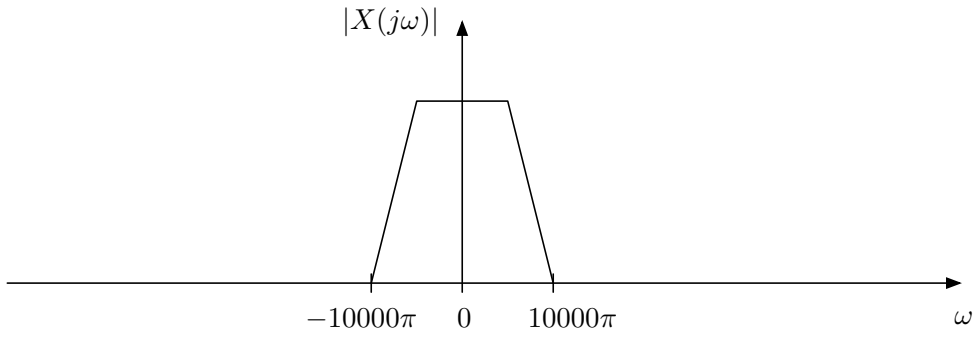


Figure 3.1

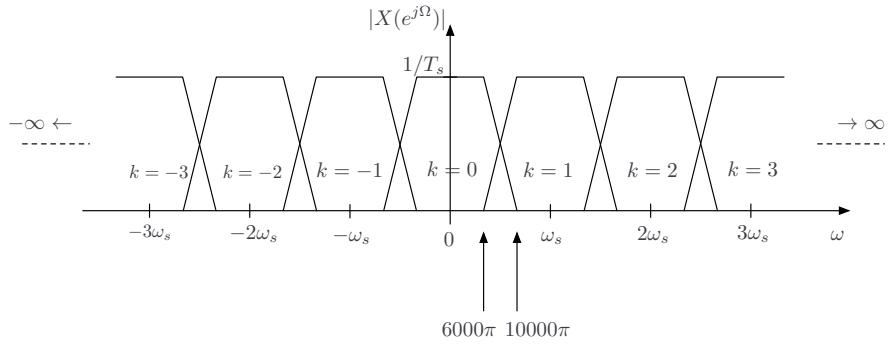
i) Given $|X(j\omega)|$ as shown and further given that the phase spectrum is odd symmetric, what property (or properties) of $x(t)$ can you deduce? [3]

ii) Now consider that $x(t)$ is sampled using an analogue-to-digital converter with sampling frequency f_s to give a resulting discrete-time signal $x[n]$ with spectrum $X(e^{j\Omega})$.

Draw a fully labelled sketch of $|X(e^{j\Omega})|$ when $f_s = 8$ kHz. [5]

iii) Describe any key relevant characteristics of $X(e^{j\Omega})$. [2]

Solution:
(i) Since the magnitude spectrum is even symmetric and the phase is odd, it can be deduced that $x(t)$ is real.
(ii) In this situation, the Nyquist sampling criteria is not satisfied so that the spectral images will overlap starting from 3 kHz.



(iii) The sampled data spectrum is aliased.

b) Consider a measurement of the acoustic impulse response between a point A and a point B of a typical room. Points A and B represent coordinate positions in the room. Assume that the acoustic propagation channel in the room can be modelled sufficiently accurately by an LTI system. An ideal loudspeaker is located at point A and it outputs an acoustic signal. An ideal measurement microphone is located at point B and it records the acoustic signal from the loudspeaker after acoustic propagation from point A to point B. The loudspeaker has an adjustable output sound

level. The microphone signal is amplified by a microphone amplifier with variable gain that feeds the signal to an analogue-to-digital converter.

In a first experiment, the loudspeaker signal is an impulse and the corresponding microphone signal is denoted $y_1[n]$.

i) What are the advantages and disadvantages of this choice of loudspeaker signal for such a measurement? [4]

ii) What effect on $y_1[n]$ would you expect to observe if the loudspeaker sound output level was too small? [3]

iii) What effect on $y_1[n]$ would you expect if the loudspeaker sound output level was too large? [3]

iv) How should the gain of the microphone amplifier be set? [3]

In a second experiment, the loudspeaker signal is white noise and the corresponding microphone signal is denoted $y_2[n]$.

v) State how the frequency response of the acoustic propagation channel could be obtained in this experiment. [2]

vi) What factor (or factors) should be taken into account when choosing the duration of $y_2[n]$? [2]

vii) Now consider that the duration of $y_2[n]$ is $N = 128$ samples and the sampling frequency employed is 48 kHz. What is the lowest frequency in Hz (not considering zero frequency) at which the frequency response of the acoustic propagation channel can be measured using this method? [3]

Solution:
(i) Advantage is that processing of the recorded signal is straightforward. The time domain impulse response is directly captured. Intuitive and easy visualisation would enable reflections to be seen and gain adjustments to be made. Disadvantage is that energy in the stimulus is very small with a high 'peak to mean' ratio. This limits the signal to noise ratio obtainable at the microphone. (Not for credit: in reality, many averages may be needed).
(ii) Noisy measurement
(iii) Clipping
(iv) Increase as much as possible without clipping so that the full dynamic range of the ADC is used.
(v) $H[k] = DFT(y[n])$
(vi) The length of $y[n]$ determines the frequency resolution of $H[k]$ obtained.
(vii) The frequency resolution is $1/NT_s$, giving 375 Hz. The DFT value of $H[0]$ will correspond to zero frequency. The value of $H[1]$ will be at 375 Hz hence this is the lowest frequency other than zero.